DetectGPT Literature Review

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Abstract

As Large Language models (LLM) become more advanced, it becomes more difficult to distinguish between human-text and LLM-generated text. While there are many fine-tuned encoder models for detecting LLM-generated text, their black-box nature makes it difficult to interpret the reasoning of detectors. This in turn makes it difficult to trust the predictions made by detectors. This survey covers supervised fine-tuned encoder models for detection that are based on BERT. We also cover post-hoc methods for interpretability such as SHAP and LIME among others. Lastly we look at models and benchmarks for detecting machine generated text across multiple languages and the constraints of these approaches.

1 Introduction

The rapid advancement of large language models (LLMs) has made it possible for LLMs to generate text that is very similar to human text. This is a cause for concern as it becomes extremely difficult to distinguish between LLM-generated text and Human written text. Currently, there are many models that aim to detect LLM text but not many of them are interpretable. This Literature review aims to highlight supervised fine-tuned models and post-hoc explainability methods to determine possible approaches to creating a model for LLM text detection that is reliable and interpretable.

2 Supervised fine-tuned encoder models

Supervised fine-tuned models are models that are pre-trained on large datasets and then fine-tuned on labeled texts that are either human or AI made. We will be focusing on models that use BERT or RoBERTa as a base.

2.1 BERT

BERT stands for Bidirectional Encoder Representations from Transformers [2]. To train BERT, Masked LM (MLM) and Next Sentence Prediction is used (NSP).

MLM is when random some random percentage of token are masked in a sentence. The hidden vectors corresponding to the tokens that were masked are fed to an output softmax over the vocabulary. 15% of the tokens of a sequence were masked at random. One downside of using MLM is that a mismatch between the pre-training and fine-tuning is created. This is because the mask token ([MASK]) does not appear during fine-tuning. To mitigate the mismatch, the mask token does not always replace the masked word.The training data generator will choose 15% of the token positions at random. If a token is selected, it will be replaced by one of three options.

(1) The mask token 80% of the time
(2) A random token 10% of the time
(3) The token is replaced 10% of the time

The token in the ith position will then be used to predict the original token using cross entropy loss.

To understand the relationship between two sentences, we pre-train the model for a binarised next sentence prediction task. This is done by taking two sentences, A and B. 50% of the time B follows A while 50% of the time sentence B is a random sentence. pre-training towards this task is beneficial to both QA and NLI.

2.2 DistilBERT

DistilBERT is a model based on BERT that has been shrunk down using knowledge distillation without receiving a major decrease in accuracy [11]. Knowledge distillation is a compression technique where a compact model is trained to reproduce the behaviour of the larger model. This model is also faster than BERT due it having less parameters.

2.3 DeBERTa

DeBERTa (Decoding enhanced BERT with disentangled attention) is a language model based on BERT that enhances its performance using disentangled attention and an enhanced masked decoder [3].

Disentangled attention: every word in DeBERTa has two vectors representing its position and content instead of just one vector. The weights of the words are computed using disentangled matrices that use the content and position of the word.

Enhanced Mask Decoder: DeBERTa uses the context and position of words when doing MLM. DeBERTa incorporates absolute word position embeddings right before the softmax layer. The model decodes the words based on the aggregated contexual embeddings of word contexts and positions.

This model is an improvement upon BERT and RoBERTa [3].

2.4 RoBERTa

RoBERTa (Robustly Optimized BERT Pretraining Approach) is a BERT based model [4] that has had the next sentence prediction removed and has been better trained on larger datasets with bigger batch sizes. This has increased its accuracy in benchmarks such as SQuAD. Like BERT, RoBERTa is a bidirectional transformer model [2]. This means that the model is able to see both the right and left side of a sentence giving it a better understanding of the context of a sentence. This makes it useful as a detector. The main disadvantage of RoBERTa based detectors is that they do not work as well on LLM text that is much different to the LLM text it is trained on.

2.5 RoBERTa-BiGRU

RoBERTa-BiGRU combines the RoBERTa model with BiGRU and uses a dual multi head attention mechanism to improve accuracy when evaluating text [13]. This model was trained on the HC3 and HC3-plus datasets. BiGRU processes text bidirectionally which helps detect subtle linguistic differences that RoBERTa struggles to detect. The multi head attention mechanism captures textual features. One of the heads determine lexical and syntactic patterns locally while the second head determines the sentence and paragraph semantics. The caveat of this model is that it was tested

on Chinese instead of English. Thus we cannot assume the model generalises well to english text.

2.6 RoBERTa fine-tuned detection using linguistic features

This model was trained using HC3-English. The HC3-English dataset was split with 90% used for training and 10% for testing. Detecting LLM-generated text was treated as a binary classification where the text is either human or machine-generated text.This model was trained specifically using linguistic features and readability metrics to determine if they can be used to further increase the accuracy of transformer based models when checking if a text is LLM generated [16]. The methods used are listed below.

Flesch Reading ease: This score aims to determine how readable a text is. This is done by calculating the average sentence length and the average amount of syllables in a word. Gunning Fog Index: This index estimates how many years of education is needed to understand a text when reading it for the first time. Perplexity: This is a measure used for language models used to test the accuracy of a probability distribution or model when predicting a sample. Burstiness: This measures the tendency of words in a text to be repeated. Word Density: This determines the amount of unique words in a text. POS Tags: POS stands for Parts-Of-Speech. These POS tags categorise words into their parts of speech to determine the syntactic structure and understand the meaning of sentences. Active voice, Passive voice and vocabulary: Vocabulary is the distinct words or tokens in the text while active and passive voice is the amount of sentences that use active or passive voice.

This training was done on multiple transformer based models with RoBERTa performing the best when tested on the test dataset. The main concern when looking at this model is that it is unclear how much of the performance increase is directly attributed to using the linguistic features versus the architecture of RoBERTa and the diversity of the training texts.

2.7 LLM-DetectAIve

LLM-DetectAIve is a tool that aims to do fine-grained detection of LLM-generated text instead of binary classification of text [1]. LLM-DetectAIve splits text into four categories. These are human-written, machine-written, machine-written, then machine-humanized and human-written, then machine-polished. Human-written text is text that is written by humans without any LLM assistance. Machine-written text is purely written by LLMs. machine-written, then machine-humanized is text that is generated by an LLM and is modified to be more human-like. human-written, then machine-polished is human-written text that is polished by an LLM.

The three models that were used to build the tool were DistilBERT, RoBERTa and DeBERTa. Each model was trained using M4GTBench. This dataset contains a mixture of human and machine written texts from many domains and generators. This dataset was split into a set of all human written text and one of machine written text. This was then further split into the 4 categories that LLM-DetectAIve uses for evaluation.

To increase the accuracy of LLM-DetectAIve, three approaches were used. These approaches were training multiple domain specific models, training one universal detector and using domain-adversarial neural network (DANN) for domain adaption. For the domain specific models, RoBERTa and DistilBERT. DeBERTa was used as the universal detector and DANN was combined with RoBERTa.

3 Post-Hoc Explainability

Many models that are used to classify text as human or machine text are black-box models. This means that we do not understand the reasoning that was used to determine the outcome. Post-Hoc explainability aims to increase interpretability of models by applying methods after training to interpret how the model works. This is different to intrinsic explainability where interpretability is built into the architecture of the model.

3.1 LIME

Local Interpretable Model-agnostic Explanations (LIME) is a post-hoc explainability technique that can explain the predictions of a classifier in an interpretable and faithful way [9]. This is important for users as this gives them understanding of the reasoning the model used. This allows users to use models more effectively. The two things a user should trust is the prediction and the model. Trusting the prediction is determining that the individual prediction is reasonable enough to act on. Trusting the model is believing that the model will behave predictably when applied in the real world.

LIME is able to give an interpretable and faithful explanation by perturbing the input to the model and learns a locally faithful linear model that explains the model locally.

3.2 DeepLIFT

DeepLIFT is defined as "a method for decomposing the output prediction of a neural network on a specific input by backpropagating the contributions of all neurons in the network to every feature of the input [12]. This is done using Equation 1 where t represents the neuron of interest, x_i .. x_n are the neurons in a layer or set layers that are required to compute t and Δt is the difference of t from the reference t_0.

$$\sum_{i=1}^{n} C_{\Delta x_i \Delta t} = \Delta t \tag{1}$$

Eq. 1 is called the summation-to-delta property.

This method is useful as it can propagates information even if the activation function is saturated which avoids placing misleading importance on bias terms. DeepLIFT can identify missed depencies by the separate treatment of positive and negative contributions.

3.3 SHAP

SHapley Additive exPlanations (SHAP) is a unified framework for interpreting predictions [5]. This framework unifies and extends six additive feature attribution methods. These are LIME, DeepLIFT, Layer-Wise Relevance Propagation, Shapley regression values, Shapley sampling values and Quantitative Input Influence.

Additive Feature attribution methods use "Explanation models", which is defined as an interpretable approximation of the original model. An attribute of the Additive attribution features is that it has

a unique solution that satisfies three criteria. Property one is local accuracy. This means that the explanation model must match the output of the original model when given a simplified version of the input that is used by the original model. Property 2 is missingness. Missingness requires that features missing in the input must have no impact. Property 3 is consistency. This states that if a simplified input's contribution increases or remains the same then the inputs attribution will not decrease.

SHAP values are values that represent the change in the model's prediction when a feature is added. These values explain how to get from the base value predicted to the current value. The calculation of these values are computationally expensive so two approximation methods have been introduces. These are Kernel SHAP (Linear LIME + Shapley values), Linear SHAP, Max SHAP and Deep SHAP (DeepLIFT + Shapley values). Kernel SHAP uses weighted linear regression to estimate the SHAP values. This is possible since Kernel SHAP uses a squared loss function and a linear explanation model. Kernel SHAP can be used for any black-box model. Linear SHAP can approximate SHAP values directly from a model's weight coefficients. The model must linear and input feature independence must be assumed. Max SHAP uses a permutation formulation of SHAP values to calculate the probability of an input increasing the maximum value of every other input. Deep SHAP approximates SHAP values by backpropagating values through the network. Deep SHAP is used for compositional models.

3.4 Other methods

This section is a brief overview of TracIn [8] and MiCE [10] [7].

3.4.1 TracIn. TracIn computes the influence of a training example on the prediction given by a model [8]. It does this by decomposing the loss of the test point after training versus at the start of training along the path the training process takes.

3.4.2 MiCE. Minimal Contrastive Editing (MiCE) is a method that produces contrastive explanations of model predictions by editting the input to change the model's prediction to the contrasting case [10].

4 Multilingual benchmarks and detection methods

These are models and benchmarks that are designed to detect LLM generated text in multiple languages. The main challenge of detectors is that they struggle to generalise to domains and languages that they are not trained for. This section will cover MULTITuDE, M4 benchmark and SemEval-2024 Task 8.

4.1 MULTITuDE

MULTITuDE is a multilingual benchmarking dataset that has human-written text and machine-generated text in 11 different languages (English, Spanish, Russian, Portuguese, Catalan, German, Dutch, Ukrainian, Czech, Arabic, and Chinese) [6]. The machine-generated text have been generated by 8 multilingual models. This benchmark was used to test zero-shot and fine-tuned models. This benchmark was also used to test models to see how well they generalised to different languages and if they improved their performance when trained on a multiple languages.

The dataset used for the 11 languages was the MassiveSumm dataset for the machine-generated text was created by taking selected articles titles and using them in prompts to 8 different language models. The dataset was split into a training and testing set. This was used to train both the monolingual and multilingual detectors.

The benchmark was used to test black-box methods, zero-shot methods and fine-tuned methods. It was found that zero-shot methods do not perform well when given multilingual data and that it is difficult to evaluate the transferability to non-English languages due to their low performance. The methods can only output one class. It was found that zero-shot methods also perform bad when used to detect English.

When testing to see if a monolingual detectors generalises to other languages, it was found that monolingual fine-tuned detectors can generalise to other languages with some degradation to performance. It was also found that the similarity of a language's linguistics mattered to generalising between languages. The performance of languages with similar linguistics correlate to each other. English was found to be an outlier language. It has a negative correlation with any other language. Detectors trained on English texts were outperformed in detecting other languages in its family by Spanish in most cases. Detectors that were trained on other languages like Spanish also performed poorly for English. It was hypothesised that the low performance was because English is the most common pre-training language for detectors and generators which could have caused the behaviour of cross-lingual capability to be different from other languages. It was also found that Non-Latin Script languages are correlated.

When testing multilingual detectors, it was found that models tested on multiple languages generalised better to unseen languages than monolingual detectors. It was also found that LLMs were able to detect text from another LLM that is from the same family (OpenAI based models do well when detecting other OpenAI models).

4.2 M4 benchmark

M4 is a large-scale benchmark that is a multi-generator, multi-domain and multi-lingual corpus [15]. This benchmark was used to test detectors across multiple generators, domains and languages.

The human-written text was taken from multiple domains (for English it was taken from Wikipedia, reddit, WikiHow, arXiv and PeerRead) and multiple languages (English, Chinese, Russian, Arabic, Urdu, Indonesian, Bulgarian). For machine-generated text, multilingual LLMs were prompted (GPT-4, ChatGPT, GPT3.5 (text-davinci-003), Cohere, Dolly-v2, and BLOOMz 176B). The LLMs were asked to write an article based on the given title, abstracts based on a paper title, peer reviews based on a paper title and abstract and news briefs based on a titles. It also summarised Wikipedia articles and answered questions posted on websites such as reddit. Multiple diverse prompts were used to create outputs with varying tones. To clean the machine-generated text, simple artifacts that help the detector classify the text were removed. The training split was 500 of human-text and machine text for development, the same for testing and the rest of the dataset for training (typically 2000 of each).

7 detectors were evaluated. These are RoBERTa, ELECTRA, XLM-R, Logistic Regression with GLTR Features, Stylistic Features (SVM based detector), NEws LAndscape (NELA) and GPTZero. 6 experiments were done using these detectors. These experiments were Same-Generator, Cross-Domain, Same-Domain, Cross-Generator, Zero-shot Evaluation: GPTZero, Multilingual Evaluation, Time Domain Evaluation and Impact of Text Length.

4.2.1 Same-Generator, Cross-Domain. For one generator, detectors were trained on data from one domain and were teted on the same domain and multiple other domains. It was found that in-domain detection is easy while out-of-domain may be hard.

4.2.2 Same-Domain, Cross-Generator. For data in one domain, detectors were trained on one generator and tested on multiple generators including itself. It was found that the highest accuracy was found when using the same generator that was used for training.

4.2.3 Zero-shot Evaluation: GPTZero. It was found that zero-shot detection for unseen generators and domains is a challenge for GPTZero.

4.2.4 Multilingual Evaluation. XLM-R was used as the detector while ChatGPT and $davinci-003$ were used as generators. This experiment was done across 7 languages. A development, test and training set was created by combining texts from across all domains. Then all the training data for the 6 other languages were combined and labeled "ALL". The data was from the same generator for the English set and the "ALL" set. The best accuracy was training and testing the detector on the same language using the same generator. When training on one generator and testing on another, The best accuracy is from the "ALL" set. XLM-R struggles to detect machine-generated text for languages that it has not seen. XLM-R still performs well for Arabic when it is trained on English.

4.2.5 Time Domain Evaluation. ChatGPT was tested across different time points to test the robustness of detectors. One detector was trained on data from March 2023 and another was trained on September 2023 data. The prompts same promts as the ones used for M4 were used and XLM-R was used as the detector. results show that a previous model may remain effective when applied to a newer generator trained on fresh data.

4.2.6 Impact of Text Length. The impact of the text length on accuracy was tested by truncating arXiv articles. The XLM-R detectors were trained. Each detector had a smaller amount of characters (1000, 500, 250, 125). These detectors were tested on data generated by ChatGPT. It was found that the accuracy of the detector decreased as the length of the text got smaller.

5 SemEval-2024 Task 8: Multidomain, Multimodel and Multilingual Machine-Generated Text Detection

This is a shared task focusing on detecting text across multiple generators domains and languages [14]. This was split into three subtasks. Subtask A is to accurately classify a text as either human or machine generated text. Subtask B is to detect if the text is human text or from a specific generator. Subtask C is determining the exact point when the author of a text goes from being considered a human to a machine. Subtask C takes into account that a text can be partially written by a human or machine.

Subtask A uses domains such as Wikipedia in its monolingual track. for its multilingual track, it works on English, Chinese, Urdu, Bulgarian, and Indonesian with each having human text and machine generated text in the dataset used. Subtask B used five generators (davinci-003, ChatGPT, Cohere, Dolly-v2, and BLOOMz) alongside human text. Subtask C used PeerRead ChatGPT generations as its training and development datasets.

For Subtask A's monolingual track, it was found that there is robust performance across methods for distinguishing between human text and machine generated text. For the multilingual track, LLMs comprehensive analysis and ensemble techniques were used to achieve better performance in deteecting human and machine generated text across multiple languages. The results of Subtask B underlines the fact that diverse and innovative approaches should be used to fine-tune models for detection. Subtask C found that combining approaches of multiple models can increase the accuracy of predictions. It was also found that it is difficult to generalise models to unseen data.

6 Conclusion

Looking at fine-tuned encoder models like RoBERTa, it is shown that while these models are good for LLM-generated text detection, they struggles to generalise to other domains and generators that it has not been trained on. Post-hoc methods such as Lime and SHAP are useful for interpretability but have not been widely used for LLM-generated text detectors. Multilingual detection is still quite challenging as detectors do not generalise across languages. The main weakness of this literature review is that the field of LLMs is moving rapidly and that the review only covers Bert derived fine-tuned encoders where there are many other approaches such as statistical approaches and zero-shot methods. There needs to be models that are accurate at detecting machine-generated text and are interpretable to protect users from false infromation.

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